Complex Analysis
Problem Set 1

In problems 1-8, give a clear geometric description of the set of complex points satisfying the given relation.

1. $|z+4| = 7$ represents a circle centered at -4 with radius 7.
2. $|z-i| < 2$ represents the interior of a circle (excluding the boundary) centered at i with radius 2.
3. $4 < |z-2i| \leq 6$ represents the region between the two concentric circles centered at $2i$ with radii 4 and 6, including the boundary of the outer circle and excluding the boundary of the inner circle.
4. $|z+2| = |z-i|$ represents the perpendicular bisector of the segment connecting -2 and i.
5. $|z-2| + |z+3i| = 5$ represents an ellipse with foci at 2 and $-3i$, with its major axis being a line segment of length 5.
6. $|z+3| - |z-1| = 2$ represents one branch of a hyperbola with foci at -3 and 1, where the difference of distances to the foci is always 2.
7. $|z+2| = \text{Re}(z) + 3$ represents a parabola with vertex at -2.5 and opening to the right, where the distance from any point on the parabola to the focus at -2 is equal to the distance from that point to the directrix at $x = -3$.
8. $|\text{Re}(z)| > 2$ represents the entire complex plane except the region with real component between $-2 \leq \text{Re}(z) \leq 2$, a vertical strip.

In problems 9-14, determine the image of the number $z = 1+2i$ under the given function and explain in your own words what the transformation does to this number. Graph both the point and its image on the complex plane.

9. $f(z) = 3z$ images the point $z = 1+2i$ to $z = 3+6i$. This transformation scales the point by a factor of 3 away from the origin, stretching the distance from the origin while maintaining the same angle.
10. $f(z) = iz$ images the point $z = 1+2i$ to $z = -2+i$. This transformation rotates the point $90°$ counterclockwise about the origin.
11. $f(z) = z+1-3i$ images the point $z = 1+2i$ to $z = 2-i$. This transformation translates the point by adding $1-3i$ to it.
12. $f(z) = (1+i)z$ images the point $z = 1+2i$ to $z = -1+3i$. This transformation scales the point by a factor of $\sqrt{2}$ and rotates it by $45°$ counterclockwise about the origin.
13. $f(z) = z^2$ images the point $z = 1+2i$ to $z = -3+4i$. This transformation squares the complex number, which results in a rotation and scaling of the point in the complex plane, squaring the modulus and doubling the argument of any given complex number.
14. $f(z) = \frac{1}{z}$ images the point $z = 1+2i$ to $z = \frac{1}{1+2i} = \frac{1-2i}{5} = \frac{1}{5} - \frac{2}{5}i$. This transformation takes the reciprocal of the complex number, which reflects the point across the real axis and inverts its distance from the origin.

Below is the complex plane illustrating the original point $z = 1+2i$ and its images under the transformations $f(z)$ in problems 9 through 14.